

뇌과학실험 I (이) (BR301-01)

학기	2026년 1학기
담당교수	조용철
이수구분	교선평수
학점	3.0
강의시간	금09:00-12:00(E4 - 225)
강의실	E4 222 (세포생물학 실험실)
소속	기초학부

교과개요

This course focuses on essential experimental techniques in neuroscience, providing hands-on experience in studying brain functions. Students will learn to design and conduct experiments, analyze data, and interpret results related to key biological mechanisms in the brain. The course emphasizes both theoretical understanding and practical skills in neuroscience research.

교과목표

Master basic neuroscience laboratory techniques. Develop skills in experimental design and data analysis. Gain a deep understanding of ethical considerations and result interpretation in neuroscience experiments.

주차별 강의계획

1주차:Introduction

2주차:Primary neuron cultures (Cortical neurons and Hippocampal neurons) 1Primary neuron cultures (Cortical neurons and Hippocampal neurons) 2

3주차:Synapse biology 1

4주차:Synapse biology 2

5주차:Human iPSC cell – neuron differentiation and organoids 1

6주차:Human iPSC cell – neuron differentiation and organoids 2

7주차:Mid-term

8주차:Astrocyte primary culture and analysis 1

9주차:Astrocyte primary culture and analysis 2

10주차:Neurological behavioral test 1

11주차:Neurological behavioral test 2

12주차:Sensory neuron culture

13주차:Axon regeneration and degeneration assay

14주차:RNA sequencing and analysis

15주차:Final-term

16주차: